

BI Dashboard Testing – Automation Strategy with Challenges & Resource Plan

Here is the enhanced BI Dashboard Testing Automation Plan with challenge analysis and resource/time estimates.

Key Implementation Challenges

Category	Challenge	Mitigation Strategy
UI Instability	Dashboard layouts may change frequently with new releases	Use locator abstraction via Page Object Model (POM)
Data Sync	DB data may change daily—causing false failures in tests	Use snapshot-based comparison / dynamic query params
Security	Role-based access may block certain tests unexpectedly	Create dedicated test roles with staged permissions
Non-standard Tools	Some BI platforms (Power BI/Looker/Tableau) don't expose APIs or identifiers easily	Use browser-based scraping carefully + iFrame handling
Query Complexity	Validation SQLs may involve complex joins or business logic	Involve Data Analyst to validate SQLs upfront
Tool Licensing	Tools like ICEDQ/DataGaps are paid solutions	Request trial POC Budget request if approved
Team Skill Gap	Team may lack automation or data testing experience	Allocate ramp-up time + provide KT sessions

Resource & Timeline Estimation

Assuming 1 QA Automation Engineer (mid-level) will be dedicated to building out the automation strategy:

Phase	Time (Est.)	Buffer	Total	Tasks
POC Setup (Tool/Code)	1.5 weeks	0.5 weeks	2 weeks	Evaluate tools, write 3–5 tests for UI & DB comparison
Framework Build	2 weeks	0.5 weeks	2.5 weeks	Selenium/TestNG base + SQL reader + Config design
Tool Integration (e.g., Jenkins, Allure)	1 week	0.5 week	1.5 weeks	Setup CI pipeline, trigger jobs, reports
Add Test Scenarios	2 weeks	1 week	3 weeks	Build 20–30 automation test cases (UI/Data)
Report & RCA Logging	0.5 weeks	0.5 weeks	1 week	Dashboard creation, failure summaries
Team KT + Maintenance Plan	1 week	0.5 week	1.5 weeks	Documentation, onboard another resource, knowledge transfer

Total Time Estimate (1 Resource):

~12-16 weeks (~3 to 4 months) with buffer time included.

Resource Profile Suggested

Profile	Experience Needed	Roles
QA Automation Engineer (Lead)	5+ years (Java, Selenium, SQL, Jenkins)	Framework setup, dashboard planning, DB query writing
BI Analyst or SME (optional)	Domain expert (data lineage, metrics)	Provide SQL validation logic, explain metric behavior
Performance Tester (optional)	JMeter/Taurus experience	Build concurrency/load testing scripts (if needed)

Optional Cost Estimate for Tools (if not open-source only)

Tool	License Model	Approx. Cost (Year)	Free Trial	Use Case
ICEDQ	Enterprise (based on rows)	\$5,000–\$15,000+		Enterprise-grade DB validation
DataGaps	Per user/project license	\$3,000–\$10,000		BI report + DB + UI validation
Wiiisdom Ops	Cloud/On-prem subscription	Varies (>\$10k typical)		Tableau/Power BI governance
JMeter/Taurus	Free	\$0		Performance testing (self-hosted)

Suggested Roadmap View (Quarter-wise)

Quarter	Milestones
Q3 2025	POC, framework base, initial dashboard coverage
Q4 2025	Expand coverage, alerting/report integration
Q1 2026	Add advanced metrics, performance tests, RCA layer